



ANALYSTLAB AFRICA

Data Science Internship Programme

Official Weekly Assignment

Week 1: Business Understanding & Data Exploration

Document Control

Programme	Data Science Internship
Week	Week 1
Project	Employee Attrition Analysis
Version	1.0
Issue Date	
Submission Deadline	Sunday, 11:59 PM (WAT)

Business Scenario

You are a Junior Data Scientist at AnalystLab Africa Consulting. Your consulting team has been engaged by ABC Manufacturing Ltd to investigate employee attrition. The HR department wants to understand why employees are leaving before investing in predictive machine learning models.

Learning Objectives

- Understand the business problem.
- Inspect and understand a real-world dataset.
- Perform initial exploratory analysis.
- Create meaningful visualisations.
- Generate business-focused insights.
- Communicate findings professionally.

Dataset

IBM HR Analytics – Employee Attrition & Performance

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Business Questions

1. What does the company's workforce look like?
2. Which departments have the highest employee attrition?
3. Does age influence attrition?
4. Does monthly income affect retention?
5. Does overtime influence attrition?
6. Which job roles experience the highest turnover?
7. Which variables appear important for future predictive modelling?

Assignment Tasks

Part 1 – Business Research

Research employee attrition, its business impact and how data science supports HR decisions. Prepare a 1–2 page Business Understanding Report.

Part 2 – Data Loading & Inspection

Load the dataset with Pandas. Inspect rows, columns, data types, missing values, duplicates and descriptive statistics. Prepare a Dataset Inspection Report.

Part 3 – Initial Data Exploration

Create at least 3 bar charts, 2 histograms, 2 boxplots and 1 pie chart. Interpret every visualisation.

Part 4 – Initial Business Insights

Provide five workforce observations and five possible factors contributing to attrition supported by evidence.

Part 5 – Reflection

Write a one-page reflection on your learning experience.

Deliverables

- Jupyter Notebook (.ipynb)
- Business Understanding Report (PDF/Word)
- Dataset Inspection Report (PDF/Word)
- Reflection Report (PDF/Word)
- Upload all files to Google Drive (Anyone with the link can view).
- Submit using the official submission form.

Assessment Criteria

Criteria	Weight
Business Understanding	10%
Data Inspection & Exploration	20%
Data Visualisation	20%
Business Insights	20%
Code Quality & Documentation	15%
Professionalism	15%

Professional Development Requirement

- Update your LinkedIn profile to reflect your role as a Data Science Intern at AnalystLab Africa.
- Publish one LinkedIn post mentioning AnalystLab Africa and using #AnalystLabAfrica.
- Publish one X (Twitter) post tagging @analystlabafri and using #AnalystLabAfrica.
- Upload your project to GitHub with a professional README.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio, demonstrates continuous learning, and increases your visibility to recruiters and potential employers.

Submission Checklist

- ☐ Business Understanding Report
- ☐ Dataset Inspection Report
- ☐ Jupyter Notebook
- ☐ Reflection Report
- ☐ GitHub Repository
- ☐ LinkedIn Post
- ☐ X Post
- ☐ Google Drive Link Submitted